AI-Based Water Quality Monitoring & Pollution Prediction — Project Proposal

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Abstract

This project builds a low-cost IoT platform that monitors water quality in real time and uses machine learning to detect pollution and forecast contamination before it becomes critical. Sensor nodes measure pH, turbidity, dissolved oxygen, temperature and TDS, and stream the data to the cloud.

AI models flag abnormal patterns and predict risk levels. A web dashboard shows quality trends across time and location, and automated alerts reach the responsible authorities or communities. The system combines software, hardware, AI/ML, IoT, cloud computing and data analytics, and belongs to the Sustainability, Climate & Green Technologies domain.

Problem Statement

Water bodies in Bangladesh, including rivers such as the Buriganga, Turag and Karnaphuli, receive industrial effluent, untreated sewage and agricultural runoff every day. Pollution spreads quickly, but detecting it is slow.

Today, water quality is checked manually: staff collect samples, send them to a laboratory and wait days for results. This approach has four gaps:

- **Slow:** contamination has already spread by the time results arrive.
- **Not continuous:** a few samples cannot capture sudden pollution events.
- **Costly:** field visits and lab tests need people, transport and equipment.
- **Reactive:** it reports what happened, and gives no warning of what is coming.

Motivation and Project Justification

We chose this project because polluted water directly harms public health and aquatic life, and an early warning can prevent much of that damage.

- **Real-world impact:** unsafe water causes diseases such as diarrhoea, typhoid and cholera, and kills fish and other aquatic life.
- **Proactive approach:** the system predicts what may happen in the coming hours or days, instead of only reporting the present state.

- **Continuous monitoring:** sensors work 24/7, so pollution events are caught as they occur.
- **Multi-technology integration:** one project brings together software, hardware, AI/ML, IoT, cloud and data analytics.
- **Low cost:** microcontrollers and affordable sensors make the system practical for a developing country.
- **Scalability:** it starts with a single pond or tank and can grow to rivers, city supply lines and factory effluent points.
- **Sustainability alignment:** it supports UN SDG 6 (Clean Water and Sanitation), SDG 13 (Climate Action) and SDG 14 (Life Below Water).

Objectives

The main objective is to design and build a real-time, AI-powered system that monitors water quality and warns of pollution early.

Specific objectives:

1. Build an IoT sensor node that measures pH, turbidity, dissolved oxygen, temperature and TDS.
2. Transmit sensor data to the cloud and store it for analysis.
3. Detect abnormal pollution patterns automatically with machine learning.
4. Forecast contamination events and risk levels ahead of time.
5. Send automated alerts to authorities or communities.
6. Provide a dashboard with trends, maps and a Water Quality Index (WQI).
7. Validate the system against real water samples and lab comparison.

Proposed Solution and Key Features

The platform covers the full path from water to decision: sense, transmit, analyse, predict, alert and visualise.

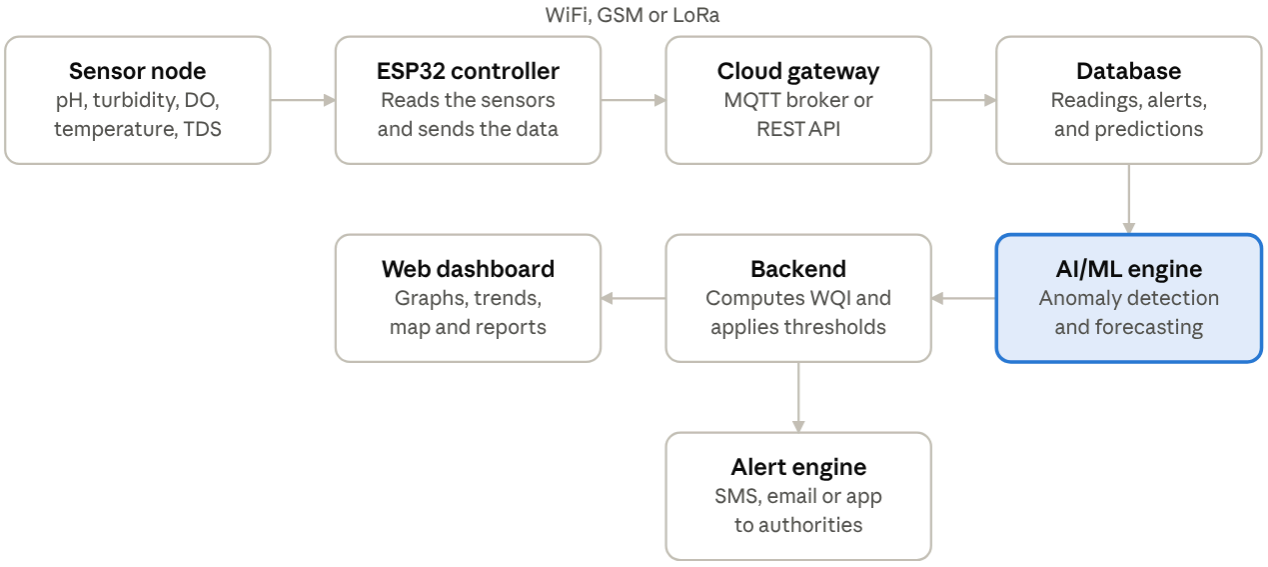
#	Feature	What it does
1	Real-time sensor monitoring	Measures pH, turbidity, dissolved oxygen, temperature and TDS continuously
2	Cloud data streaming	Sends readings to the cloud over WiFi, GSM or LoRa
3	Anomaly detection (AI)	Detects abnormal pollution patterns as they occur

#	Feature	What it does
4	Pollution forecasting (ML)	Predicts possible contamination events before they turn critical
5	Automated alerts	Notifies authorities or communities by SMS, email or app
6	Dashboard	Shows graphs, trends and a map by time and location
7	Water Quality Index	Combines all parameters into one score: Good, Moderate or Poor
8	History and reports	Stores past data and exports reports as PDF or CSV
9	Multi-location support	Handles many sensor nodes, each shown on the map
10	User roles	Admin, Authority and Public viewer accounts

Optional extras if time allows: GPS pollution heatmap, mobile app or PWA, solar-powered nodes, Bangla and English interface, public complaint option and a data API for researchers.

System Architecture

Sensor data flows through AI to alerts and the dashboard



system architecture · 8 components, 7 links

Readings travel from the sensor node through the cloud into the database; the AI engine and backend then turn them into dashboard views and alerts.

Technology Stack

We use affordable, widely available tools so the system stays low-cost and easy to reproduce.

Layer	Technology
Microcontroller	ESP32 (built-in WiFi)
Sensors	pH, turbidity, dissolved oxygen, DS18B20 (temperature), TDS
Communication	WiFi, optional GSM (SIM800L) or LoRa for remote areas
Power	Solar panel with battery
Messaging	MQTT (Mosquitto or HiveMQ) or HTTP REST
Backend	Python (Flask or FastAPI) or PHP
Database	MySQL, InfluxDB or Firebase
AI/ML	Python, scikit-learn, TensorFlow/Keras, pandas
Cloud	Firebase, AWS IoT, Google Cloud or a free-tier VPS
Frontend	HTML, CSS, JavaScript, Chart.js or Plotly, Leaflet.js for maps

AI/ML Module

The AI layer does two jobs: it spots pollution as it happens, and it predicts what comes next.

Task	Candidate models	Purpose
Anomaly detection	Isolation Forest, One-Class SVM, Autoencoder	Flag abnormal readings in real time
Forecasting	ARIMA, Prophet, Random Forest, XGBoost, LSTM	Predict WQI and contamination risk

We will train on a public water quality dataset, extend it with our own sensor readings, and add synthetic data where samples are scarce. Simple baselines (ARIMA, Isolation Forest) come first; the more complex models are kept only if they perform better.

The Water Quality Index combines all parameters into a single score using a weighted average:

```
WQI = \frac{\sum_{i=1}^{n} q_i w_i}{\sum_{i=1}^{n} w_i}
```

Here q_i is the quality rating of parameter i and w_i is its weight. The score maps to three levels: Good, Moderate and Poor.

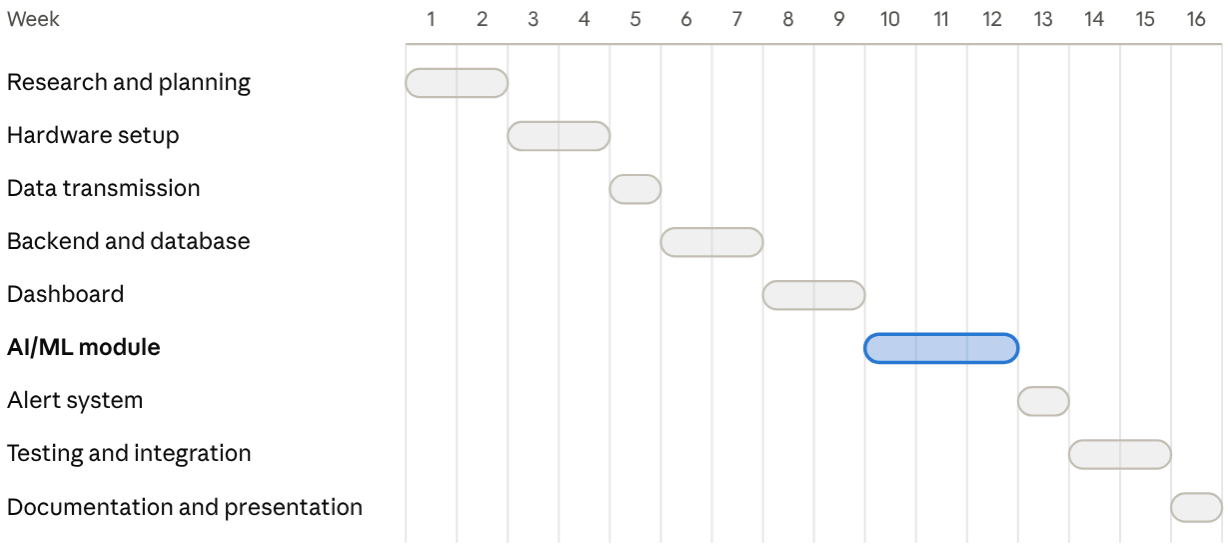
Database Design

Six tables hold users, locations, devices, readings, predictions and alerts.

Table	Key fields	Purpose
users	id, name, role, email, password	Accounts and access roles
locations	id, name, latitude, longitude, water_body_type	Where monitoring happens
sensor_nodes	id, location_id, status, last_seen	Each deployed device
sensor_readings	id, node_id, ph, turbidity, do, temp, tds, timestamp	Raw measurements
predictions	id, node_id, predicted_wqi, risk_level, forecast_time	AI output
alerts	id, node_id, type, message, sent_to, timestamp	Alert history

Development Plan and Timeline

The 16-week plan builds hardware first and AI/ML in weeks 10 to 12



16-week plan · 9 phases

Hardware and data flow come first, AI/ML runs in weeks 10 to 12, and weeks 14 to 16 cover testing and documentation.

Evaluation Plan

We will prove the system works by testing the sensors, the models and the full pipeline.

What we test	How we measure it
Sensor accuracy	Compare readings with lab results on the same water samples
Anomaly detection	Precision, recall and F1-score
Forecasting	RMSE against actual later readings
Alert speed	Time from a pollution event to the alert reaching the user
Reliability	System uptime during the test period
Cost	Manual sampling cost compared with the automated system

Final validation uses real water samples: clean tap water, pond water and deliberately contaminated water in a bucket test.

Challenges and Mitigation

Challenge	Planned solution
Sensor drift and inaccuracy	Regular calibration against known solutions
No internet in remote areas	GSM or LoRa communication
Limited training data	Public datasets plus synthetic data augmentation
Power supply outdoors	Solar panel with battery
False alarms	Combine AI output with multiple thresholds before alerting
Water damage to electronics	Sealed waterproof enclosure

Estimated Budget

One sensor node costs roughly BDT 21,000 to 35,000. These are approximate figures from memory and should be checked against current local market prices.

Item	Approx. cost (BDT)
ESP32 board	600 to 900
pH sensor and module	2,500 to 4,500
Turbidity sensor	600 to 1,200
Dissolved oxygen sensor	12,000 to 20,000
TDS sensor	600 to 1,000
DS18B20 waterproof temperature sensor	250 to 400
Solar panel, battery and charger	2,500 to 4,000
GSM module (optional)	700 to 1,000
Enclosure, wiring and misc.	1,500 to 2,500
Cloud hosting (free tier)	0

Item	Approx. cost (BDT)
Total per node	**about 21,250 to 35,500**

The dissolved oxygen sensor is the largest cost and can be added in a later phase if the budget is tight.

Expected Outcomes and Impact

The finished project delivers a working prototype and a clear case for wider use.

- A working sensor node that streams live water quality data to the cloud.
- Trained models that detect pollution anomalies and forecast risk levels.
- A dashboard and alert system that cut the delay between pollution and response from days to minutes.
- Faster pollution control, better protection of public health and aquatic ecosystems.
- A design that adapts to different water bodies and monitoring contexts.

Future Scope

- GPS-based pollution heatmap across many nodes.
- Android app or progressive web app for the public.
- Solar-powered, LoRa-connected nodes for areas without WiFi.
- Pollution source tracing, estimating which direction contamination comes from.
- Public complaint and reporting option.
- Bangla and English interface.
- Open data API for researchers.

Conclusion

Manual water testing is too slow to stop pollution from spreading. This project replaces it with continuous IoT sensing, AI-based detection and forecasting, and automated alerts. It addresses a real environmental problem, uses a broad set of modern technologies, and can grow from a single prototype to a network of monitoring stations.